Evolution of Natural Language Processing : From Rule-Based Systems to Modern AI

DSA0311 – Natural Language Processing

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Introduction

- NLP stands for Natural Language Processing.
- It is a branch of Artificial Intelligence (AI).
- It helps computers understand and process human languages.
- NLP is used in chatbots, translation, voice assistants, and text analysis.

Example: Google Translate converts text from one language to another.

Early NLP – Rule-Based Systems

- Early NLP systems used manually written rules.
- Linguists and programmers created grammar and vocabulary rules.
- Systems followed fixed instructions to process sentences.
- They worked well for simple tasks but struggled with complex language.

Example: A chatbot replies “Hello” when the user types “Hi.”

Limitation: It cannot easily understand sentences that are different from its predefined rules.

Statistical NLP and Machine Learning

- NLP started using statistics and data to understand language patterns.
- Machine learning models learned from examples instead of depending only on fixed rules.
- They improved tasks such as spam detection and speech recognition.
- Their performance depended on the quality and amount of training data.

Example: An email system learns to identify spam messages.

Deep Learning in NLP

- Deep learning uses artificial neural networks.
- Models learn complex patterns in words and sentences.
- Word embeddings represent words as numerical vectors.
- Recurrent Neural Networks (RNNs) and LSTMs helped process sequences of text.

Example: A translation system uses the context of a sentence to produce a better translation.

Transformers and Modern NLP

- The Transformer architecture was introduced in 2017.
- It uses an attention mechanism to identify relationships between words.
- Large Language Models (LLMs) are trained on large amounts of text.
- Modern NLP can generate text, answer questions, summarize documents, and assist with coding.

Examples: ChatGPT, BERT, and Google Translate.

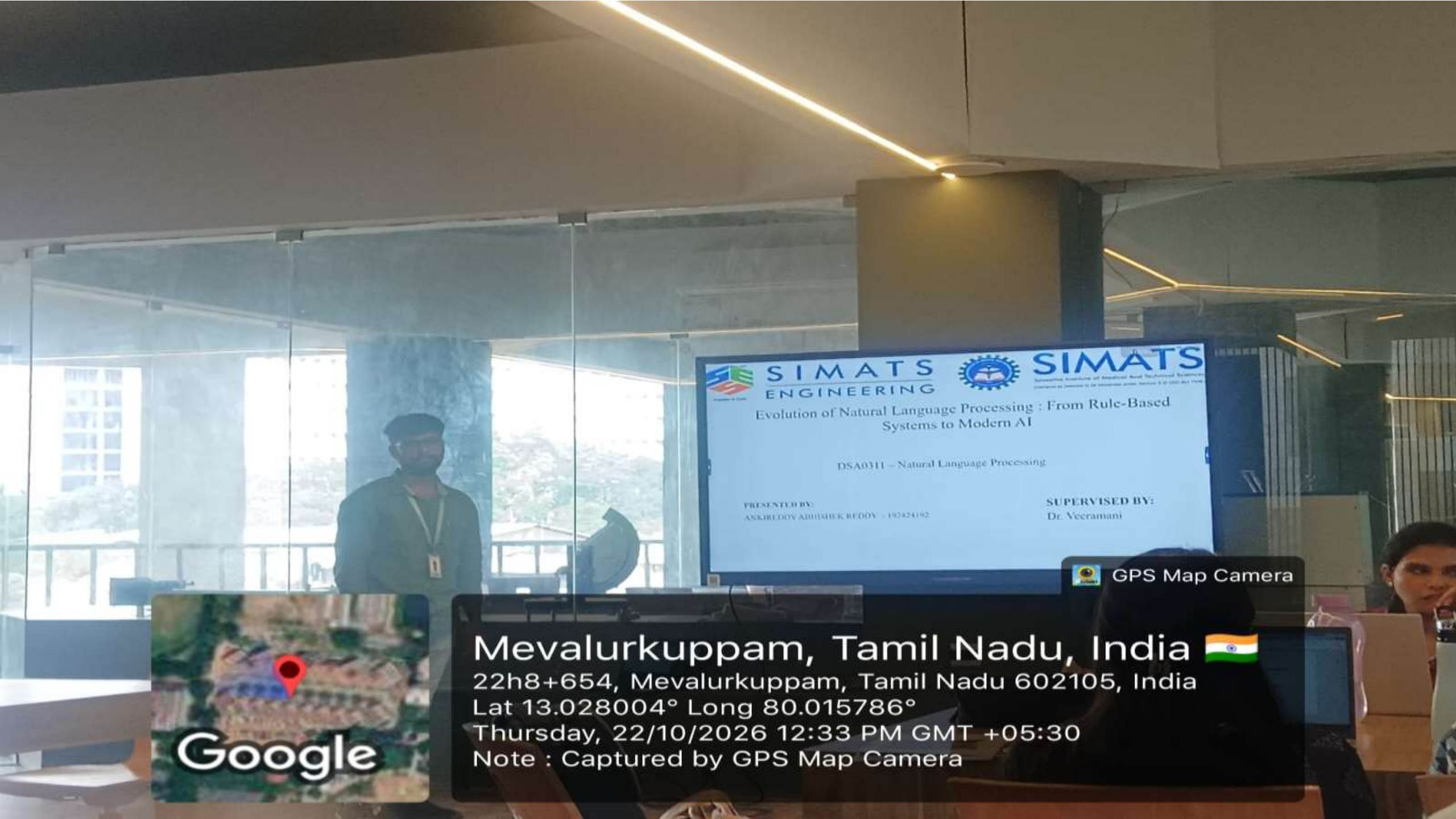
How to explain: “Transformers made a major improvement in NLP. Modern models can understand context and generate human-like responses, although they can still make mistakes.”

Applications and Conclusion

- Chatbots: Answer user questions.
- Translation: Convert one language into another.
- Healthcare: Help analyze medical documents.
- Education: Support learning and writing.
- Business: Analyze customer feedback and reviews.

Conclusion

- NLP evolved from fixed rules to machine learning, deep learning, and modern language models.
- These advances have made language processing more flexible and powerful.
- The future of NLP focuses on better accuracy, multilingual support, and responsible AI.


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